

SUNDARA RAO BADAGALA

Software Engineer (Frontend)

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[Linkedin](#) | [Github](#) | [Portfolio](#)

SUMMARY

Software engineer with 4 years and 10 months specializing in edtech platforms, building scalable React.js and Next.js applications used by 500K+ concurrent users. Experienced in designing feature-based modular architecture, enabling scalable, maintainable, and efficient frontend systems. Strong focus on reusable components, performance optimization, and delivering responsive, user-centric interfaces in fast-paced environments.

EXPERIENCE

Software Engineer — Greater Than Educational Technologies

Jun 2022 – Jun 2026

- Architected and developed scalable web applications using **React.js, Next.js, Node.js, and TypeScript**, improving maintainability and reducing code complexity by 30%
- Built feature-based modular frontend architecture with reusable components and optimized folder structures for scalable development
- Designed reusable component architecture and optimized rendering strategies for applications supporting high concurrent traffic
- Implemented **SSR, lazy loading, code splitting, and performance optimization techniques**, improving Core Web Vitals and reducing initial page load times
- Integrated and managed secure **RESTful/JSON APIs** for learning platforms, payment systems, and proctoring workflows
- Maintained clean, reusable, and well-documented code using **Git version control** and agile development workflows

Frontend Developer — Nspira

Aug 2021 – May 2022

- Built responsive and cross-browser compatible user interfaces with optimized API handling and state management
 - Worked extensively with **REST APIs, Axios, Redux Toolkit, and Context API** for scalable state and data management
 - Implemented unit testing using Jest for frontend components, ensuring stable feature releases and improving overall application reliability
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PROJECTS

Coaching Centers

Technologies: [NextJS, TypeScript, Tailwind, Recoil]

- Owned the application end-to-end, handling architecture, feature development, API integration, optimization, and production support

nLearn: [Online Tech Ed Platform]

Technologies: [ReactJS, JavaScript, Redux, Axios, Styled Components, Bootstrap]

- Implemented “Astra” AI chatbot using React.js/Next.js with Server-Sent Events (SSE) for real-time streaming responses integrated with AI-powered conversational APIs
- Built an interactive online drawing module by converting PDFs into drawable image canvases, enabling students to write/draw and submit responses digitally, increasing student engagement by 20%.
- Built a Listening Comprehension feature with custom audio player integration, enabling students to attend audio-based assessments and submit responses digitally
- Integrated Mixpanel to track student activities, analyzing the most and least visited modules. Leveraged insights to enhance features, resulting in a 20% increase in student engagement.

nMonitor: [Student Exam Monitoring System]

Technologies: [ReactJS, Redux, Material UI, Axios]

- Developed a real-time student exam activity monitoring system, reducing teachers' workload by 40% by automating exam supervision.
- Led the development and maintenance of the application, ensuring scalability and reliability.

CNAPS: [Online Exam Olympiad Platform]

Technologies: [ReactJS, Redux Toolkit, Ant Design, Axios]

- Designed and developed a proctoring feature to track student activities during exams by capturing snapshots and analyzing sound levels to detect potential cheating, enhancing exam security by 40%.
- Implemented Razorpay payment gateway, increasing app usage by 30%.

nCMS

Technologies: [ReactJS, TypeScript, Redux Toolkit, Material UI, Axios]

- Implemented scalable dynamic forms using Formik hooks and optimized large dataset rendering through window virtualization techniques

TECHNICAL SKILLS

Frontend Technologies: React.js, Next.js, JavaScript (ES6+), TypeScript, HTML5, CSS3

State Management: Redux Toolkit, Recoil, Context API

Styling Libraries: Tailwind CSS, Material UI, Chakra UI, Bootstrap, Styled Components

Backend Technologies: Node.js, Express.js, MongoDB

Others: Git, Axios, JIRA, Mixpanel, REST APIs, Storybook, Jest, SEO Fundamentals

EDUCATION

B.Tech | 2017 | CGPA: 7.5/10

Rajiv Gandhi University of Knowledge Technologies (RGUKT), Nuzvid